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Ex - 2

Study Of Network Cables.

Aim:- To understand different types of network cables.

a) Different types of cables used in networking are

1) Unshielded twisted pair (UTP) cable
2) Sheilded twisted pair (STP) cable
3) Coaxial cable
4) Fibre optic cable

Cable type	Category	Maximum data transmissia	Advantages/ disadvantages	Application/ use	Image
UTP	Category 3	10 bps	Advantage → Cheap, easy to install	10 base-T Ethernet	
	Category 5	upto 100 mbps	Disadvantage → More prone to electromagnetic interference, noise	Fast ethernet Gigabit ethernet	
	Category 5e	1 Gbps			
STP	Category 6, 6a	10 Gbps	Advantage → Sheilded, faster than UTP.	Gigabit Ethernet, 10G (55m) used in data centres	
SSTP	Category 7	10 Gbps	Disadvantage → Expensive and Greater installation effort		
Coaxial cable	RG-6, RG-59 RG-4	10-100 Mbps	Advantage → High bandwidth → Immune to interference Disadvantage → limited distance, cost	Television network	

fibre optics table	Single mode multi mode	100Gbps	Advantages → High speed, bandwidth, security Disadvantage → Costly, skilled installers required	Maximum distance of fibre optics cable is around 100 m.



b) Make your own ethernet cross-over cable / straight cable

Tools required:

→ Ethernet cabling. CAT5e is required for gigabit support, but CAT5 cabling works as well just over short distances.

→ A chrimping tool, Two RJ45 plugs, Two plug sheilds

Step 1:-

To start construction of the device, begin by threading sheilds onto the cable.

Step 2:-

Next, strip approximately 1.5 cm of cable sheilding from both the ends. The chrimping tool has a round area to complete the task.

Step 3:-

After, you will need to untangle the wires. There should be four 'twisted pairs'. Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4:-

Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you as shown.

Step 5:-

Next, push the cable right in. The notch at the end of the plug needs to be just over the cable sheilding; and if it isn't, that means you stripped off too much sheilding. Simply snip the cables back a little more.

Step 6:-

After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7:-

Lastly, repeat for the other end using diagram B (to make crossover cable)/ using diagram A (to make straight through cable)

Student observation

Q) ~~What is the~~

→ Straight cables connect dissimilar devices by keeping the pin order identical on both ends.
Crossover cables connect similar devices by swapping the transmit and receive wire pairs internally.

→ A crossover cable is used to connect two PCs directly to each other.

→ A straight cable is used to connect a router or a switch to your PC.

RESULT:-

~~This~~ different types of cable networks are learned successfully.